There are several techniques to combine the signal and carrier wave. One such technique is amplitude modulation where the amplitude of the carrier wave is varied according to the signal. On the other hand, the frequency of the wave is varied according to the signal in the case of frequency modulation (FM).

2.4 The transmitter and receiver

A transmitter is an electronic device, which, with the help of an antenna, sends an electromagnetic signal such as a radio wave, either to a receiver or a device that will retransmit the signal to a receiver.

A basic transmitter consists of a power supply, an oscillator, an amplifier and a modulator. There are various types of transmitters depending on the varied number of applications. In many parts of the world, the use of a transmitter is controlled by law as it can interfere with waves used for other purposes such as aircraft navigation, and hence endanger lives.

The receiver is an electronic device that receives the transmitted wave from the transmitter. A current is induced in the receiver's antenna by the incoming wave and this regenerates the original signal. The signal is then amplified to drive a reproducing device such as a loudspeaker, tape recorder, earphone or video monitor. The mechanism of reception is in the electromagnetic waves inducing voltages on the metallic antenna which are amplified. Selection is the process of selecting one frequency range from the signal. The better the receiver is at differentiating between the desired and undesired frequencies, the better the selectivity rating. The sensitivity of a receiver is the ability to detect modulated waves from noise. The receiver must have some mechanism to demodulate the signal that was initially modulated at the transmitter. Therefore, it must separate the high frequency carrier wave from the information that was sent.

Probably the simplest of all methods of demodulation is the demodulation of an amplitude modulated carrier wave. It usually consists of a single diode and filter. Other demodulation techniques include:

- DSB and SSB demodulation
- BFM demodulation